

Coordination of Water Flow with Optimal Power System Operation

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Abstract: Energy and water are essential resources for modern society. Their increasing demands result in various social, economic, and environmental issues. While traditional approaches consider energy and water distribution systems as two uncouple systems, they are interdependent in reality, and this relationship is often known as the water-energy nexus. This paper proposes a new model to optimize both the water distribution network and the energy network which runs the water distribution system. Additionally, we propose a fully distributed algorithm based on the Alternating Direction Method of Multiplier (ADMM), which eliminates the need for a central coordinator and enhances the privacy and security of the system. The effectiveness of the proposed optimization scheme is verified by numerical simulations.

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Keywords: Multi-agent systems, Water distribution system, ADMM, distributed optimization.

1. INTRODUCTION

Water and power network are traditionally considered as separate systems Macknick et al. (2011); Ahn et al. (2018); Van Nguyen et al. (2019); Sun et al. (2021). However, as the demand for energy and water increases, the interdependence between these two systems has gained attention recently Oikonomou et al. (2017), Santhosh et al. (2013). Water plays a vital role in energy generation, such as hydroelectric, power plant cooling, and thermal-power generation facilities. On the other hand, electricity is an essential requirement for the water system, where it can be utilized for water treatment in the water supply plants and the operation of pumps in the water distribution system. The interdependency between water and energy is often referred to as the water-energy nexus Oikonomou et al. (2017). From the modeling perspective, the water-energy nexus problem is usually formulated as a multi-objective optimization problem that combines both water and energy objectives Kurek and Ostfeld (2013). The work El-Nashar (2001) considered several options for power and distillation co-generation and proposed a methodology for selecting the optimal co-generation option based on both technical and economic viewpoints. Water consumption for electricity generation has been investigated in Macknick et al. (2011), they estimated water consumption for energy generation in the US. The study conducted by Jin et al. (2019) involved the collection of data on water consumption during different life cycle stages of power generation. They also show that different power types may require a different amount of water consumption. An optimization model to minimize water consumption

for a cooling system of the Natural Gas Combined Cycle power plant was developed in Jin et al. (2019). The authors in Santhosh et al. (2014b) developed the multi-plan real-time economic dispatch of both power and water, while in Santhosh et al. (2014a), they investigated the impact of water and electricity storage on water-energy production.

Conventionally, water distribution systems are usually considered as a single entity that provides water to a whole community of users. However, in reality, there usually exist multiple water supply parties responsible for providing water to a city. As a result, existing works require a central computational unit to collect information from the entire water network, which may raise privacy concerns since the water supply parties can be private companies. In addition, a failure on the central computational unit can potentially disable the whole system.

To overcome these aforementioned drawbacks, we present a novel approach for the water-energy nexus problem at the demand side with a multi-communities structure. Our model enables multiple water supply parties to cooperatively provide water to a whole city, with each party responsible for a specific community within the city. The allocation of water for the water supply plants and water loads is assigned to minimize the costs of water generation and maximize users' satisfaction levels. Based on the assigned water generations and loads, the pumps are operated efficiently in order to minimize the cost for energy generation. Next, we propose a fully distributed optimization algorithm based on the ADMM approach Boyd et al. (2011) to solve the formulated problems, thereby enhancing the privacy of the system and sharing the computational burdens with distributed computational units Nguyen et al. (2022). We demonstrate that in each community, the water supply party only needs to communicate with

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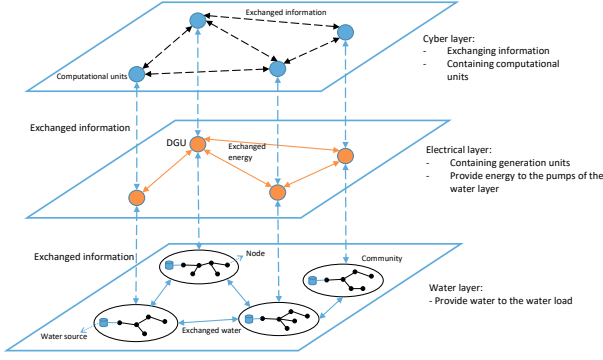


Fig. 1. Illustration of Water-Energy network

their neighbors and reveal unimportant information, then the privacy of the system is improved.

The remaining parts of this paper are outlined as follows. We first introduce the structure of the water-energy network in Section 2. In this section, we also formulate two problems for water and energy optimization. Next, in Section 3, we propose distributed optimization algorithms for the two formulated problems. Section 4 includes numerical simulations to illustrate the effectiveness of the proposed algorithm. Finally, Section 5 gives the conclusion for this paper.

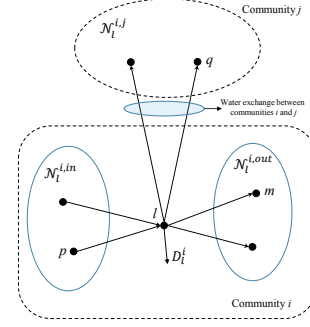
2. PROBLEM FORMULATION

Consider a network of n communities (or agents). Each community is equipped with one water source (such as a water supply plant or reservoir) and one distributed generation unit (DGU) for electricity supply (as in Fig.1). In each community, the DGU provides energy to carry water from the local source to the water load across the whole community. Let the undirected graph $\mathcal{G}(\mathcal{V}, \mathcal{E}^W, \mathcal{E}^E)$ represent the network of n communities, where $\mathcal{V}$ denotes the set of communities, $\mathcal{E}^W, \mathcal{E}^E$ denote the set of water pipes and electricity lines between the communities, respectively. Since each community contains one water source and one DGU, $\mathcal{V}$ is also the set of water sources and DGUs. Water and electricity can be exchanged among communities through $\mathcal{E}^W$ and $\mathcal{E}^E$ in case of water and electricity surplus or deficit. Let $\mathcal{N}_i^W = \{j | (i, j) \in \mathcal{E}^W\}$ denote the set of neighboring agents of agent i in terms of the water layer, and $\mathcal{N}_i^E = \{j | (i, j) \in \mathcal{E}^E\}$ denote the set of neighboring agents of agent i in terms of the electrical layer.

Each community is a sub-network consisting of pipes, junctions, pumps, and tanks. Junctions and tanks are the nodes of the sub-network. In each community i , if there exists a pipe between node l and node m , we say that l and m are neighbors. Let us denote $\mathcal{N}_l^i$ as the set of neighboring nodes of node l in the community i . We focus on the time slot size δ , the set of time slots is denoted by $\mathcal{K} = \{1, 2, \dots, K\}$, and the time index is represented by $t \in \mathcal{K}$.

2.1 Modeling of the Water Distribution System

Next, we present the constraints at each element of the network.


 Fig. 2. Water flows at node l in community i

Junctions Consider a community i and one of its neighbors $j \in \mathcal{N}_i$. Let $\mathcal{N}_l^{ij}$ be the set of nodes of community $j \in \mathcal{N}_i^W$ such that they are connected to junction l inside the community i by pipes. At the junction l , we have the following constraint based on water conservation rule

$$\sum_{p \in \mathcal{N}_l^{i, in}} Q_{pl}^i(t) = \sum_{m \in \mathcal{N}_l^{i, out}} Q_{lm}^i(t) + D_l^i(t) + \sum_{j \in \mathcal{N}_i^W} \sum_{p \in \mathcal{N}_l^{ij}} Q_{lp}^{ij}, \quad (1)$$

where, $Q_{pl}^i(t), Q_{lm}^{ij}(t)$ denote the water flow from node p to node l in community i , and between communities i and j , respectively; $D_l^i(t)$ denotes the water load at node l in community i . For water balance, for any junction l at community i , we have the following constraints

$$Q_{lq}^{ij} + Q_{ql}^{ji} = 0, \forall i \in \mathcal{V}, j \in \mathcal{N}_i^W, q \in \mathcal{N}_l^{ij}. \quad (2)$$

Notice that, if $q \in \mathcal{N}_l^{ij}$ then we also have $l \in \mathcal{N}_q^{ji}$.

Let $H_l^i(t)$ represent the total head at junction l or at the outlet of tank l in community i . Denote the minimum and maximum total heads at junction l in community i by $H_l^{i, min}$ and $H_l^{i, max}$, we have the following constraints

$$H_l^{i, min} \leq H_l^i(t) \leq H_l^{i, max}. \quad (3)$$

Water sources Each community has a water source, which can be a water supply plant that serves its local demand and possibly exchanges water with other communities. The total head at each local source is assumed to be constant Fooladivanda and Taylor (2018), i.e., $H_s^i(t) = H_s^i = \text{const}$. The following constraint applies to the water generation $Q_G^i(t)$ at the source of community i as

$$0 \leq Q_G^i(t) \leq Q_G^{i, max}. \quad (4)$$

Pipes Let $H_{lm}^{i, Loss}(t)$ be the loss of head through pipe $(l, m) \in \mathcal{P}_i$, with $\mathcal{P}_i$ denotes the set of pipes in community i , which can be computed by the Darcy-Weisbach equation Fooladivanda and Taylor (2018) as follows

$$H_{lm}^{i, Loss}(t) = f_{lm}^i \times [Q_{lm}^i(t)]^2, \quad (5)$$

where f_{lm}^i denotes the loss coefficient of pipe (l, m) , which depends on the pipe's property. If the pipe (l, m) does not contain a pump, we have

$$H_l^i(t) = H_l^i(t) + H_{lm}^{i, Loss}(t). \quad (6)$$

Pumps If the pipe $(l, m) \in \mathcal{P}_i^u$ contains a pump (l, m) , with $\mathcal{P}_i^u$ denotes the set of pumps in community i , then

the pump increases the pressure head by the amount of $H_{lm}^{i,G}(t)$ as

$$H_l^i(t) + H_{lm}^{i,G}(t) = H_m^i(t) + H_{lm}^{i,Loss}(t). \quad (7)$$

As we will show later, pumps are required to increase the head pressure through the pipes to meet the demand for the water network. For operating, pumps consume electricity; thus, pumps can be considered as the element connecting water and energy network.

Tanks Let $V_l^i(t)$ denote the water volume in tank l in community i , the constraint for water conservation in tank $l \in \mathcal{T}_i$, with $\mathcal{T}_i$ being the set of tanks in community i , as

$$V_l^i(t) = V_l^i(t-1) + \left(\sum_{p \in \mathcal{N}_l^{i,in}} Q_{pl}^i(t) - \sum_{m \in \mathcal{N}_l^{i,out}} Q_{lm}^i(t) \right), \quad (8)$$

$$V_l^{i,min} \leq V_l^i(t) \leq V_l^{i,max}, \quad (9)$$

$$Q_{pl}^i(t) \geq 0, p \in \mathcal{N}_l^{i,in} \text{ and } Q_{lm}^i(t) \geq 0, m \in \mathcal{N}_l^{i,out}, \quad (10)$$

where $V_l^{i,min}$ and $V_l^{i,max}$ are the volume limitations of the tank. Unlike any junctions, there are two types of the pressure head of a tank: the head at the inlet of the tank $H_{(in)l}^i(t)$ and the head at the outlet of the tank. Note that the head at the outlet of a tank is also the total head of the tank $H_l^i(t)$

$$H_{(in)l}^i(t) \geq H_l^i(t). \quad (11)$$

Let A_l^i denote the cross-sectional area of tank l , and denote e_l^i the elevation of tank l . The total head of tank l is the summation of the elevation and the water level of the tank as follows

$$H_l^i(t) = V_l^i(t)/A_l^i + e_l^i. \quad (12)$$

2.2 Electrical Network

The electrical network supplies energy to the pumps to distribute water across the water network. Assume that each community is equipped with a distributed generation unit (DGU), which provides electricity to its local pumps.

Distributed Generation Units We consider the quadratic cost function for energy generation at DGU_i Ahn et al. (2018) as follows

$$E_i(P_i(t)) = \alpha_i(t) (P_i(t))^2 + \beta_i(t) P_i(t) + \gamma_i(t),$$

where $P_i(t)$ is the amount of energy generated by DGU_i , and $\alpha_i(t), \beta_i(t), \gamma_i(t)$ are positive parameters which depend on the property of DGU_i , and can vary over time.

Let $P_{ij}(t)$ be the amount of transmitted energy from community i to community j , and $P_{Li}(t)$ are the total energy used by all pumps in community i , in time slot t . We have

$$P_i(t) = P_{Li}(t) + \sum_{j \in \mathcal{N}_i^E} P_{ij}(t), \quad (13)$$

$$P_i^{min} \leq P_i(t) \leq P_i^{max}, \quad (14)$$

where P_i^{min} and P_i^{max} denote the limitations of DGU_i . For energy balance, we have the following constraints

$$P_{ij}(t) + P_{ji}(t) = 0, \forall (i, j) \in \mathcal{E}^E. \quad (15)$$

Energy Consumption of Pumps Consider a pump $(l, m) \in \mathcal{P}_i^u$. The energy consumption of pump (l, m) in time slot t is computed as Fooladivanda and Taylor (2018)

$$E_{lm}^i(t) = \frac{\rho g}{h_{lm}^i} H_{lm}^{i,G}(t) Q_{lm}^i(t), \quad (16)$$

$$E_{lm}^{i,min}(t) \leq E_{lm}^i(t) \leq E_{lm}^{i,max}(t), \quad (17)$$

where ρ is the density of water, g is the local gravitational acceleration, and h_{lm}^i denotes the efficiency of pump (l, m) . The capacity of pump (l, m) is denoted by $E_{lm}^{i,min}(t)$ and $E_{lm}^{i,max}(t)$. Since $P_{Li}(t)$ is the total energy utilized by all pumps in community i , one has

$$P_{Li}(t) = \sum_{(l,m) \in \mathcal{P}_i^u} E_{lm}^i(t). \quad (18)$$

2.3 Optimization Problems

Coordination of Water Flow Our purpose is to minimize the water generation cost at local sources and maximize the satisfaction levels at junctions. To formulate the cost of producing water, we assume that the cost includes two components: a fixed cost and a variable cost. The water generation cost can be written as $C_i(Q_i^G(t)) = a_i(t) (Q_i^G(t))^2 + b_i(t) Q_i^G(t) + c_i$. Thus, the objective function for water coordination can be formulated as follows

$$J_W = \sum_{t \in \mathcal{K}} \sum_{i \in \mathcal{V}} \left[C_i(Q_i^G(t)) - \sum_{l \in \mathcal{J}^i} U_l^i(D_l^i(t)) \right], \quad (19)$$

where, $U_l^i(D_l^i(t))$ is the utility function of water use at junction l inside the community i , and is defined in a parabolic form as follows Jia et al. (2014)

$$U_l^i(D_l^i(t)) = \varepsilon_l D_l^i(t) - \delta_l (D_l^i(t))^2. \quad (20)$$

We have the following problem

$$\mathbf{P1} \quad \min \quad J_W = \sum_{i \in \mathcal{V}} J_{Wi}, \quad (21a)$$

$$\text{s.t.} \quad (1), (2), (4), (8) - (10), \quad (21b)$$

$$\text{where } J_{Wi} = \sum_{t \in \mathcal{K}} \left[C_i(Q_i^G(t)) - \sum_{l \in \mathcal{J}^i} U_l^i(D_l^i(t)) \right].$$

Problem **P1** aims to determine the water generation at each water source, water flows at each pipe, water loads at junctions, and the volumes of tanks.

Coordination of Energy Flow At the electrical layer, our purpose is to minimize the cost of total energy generation $J_E = \sum_{i \in \mathcal{V}} E_i(P_i(t))$.

For energy coordination, we focus on finding the energy generations, flows of the electrical network, the pressure at each node, and the head gains of pumps to minimize the cost of power generation while satisfying the water coordination obtained from **P1**. We have the problem **P2** as

$$\mathbf{P2} \quad \min \quad J_E = \sum_{i \in \mathcal{V}} E_i(P_i(t)), \quad (22a)$$

$$\text{s.t.} \quad (3) - (7), (11), (12), (13) - (18). \quad (22b)$$

3. DISTRIBUTED SOLUTION TO P1 AND P2

In this section, we provide the distributed algorithm based on ADMM Boyd et al. (2011) to solve problem **P1** and **P2**.

3.1 Solution to **P1**

For convenience, let us stack all the variables of agent i related to water flows in **P1** to $X_i(t)$. Denote $X_i = \{X_i(t)\}_{t \in \mathcal{K}}$ and $X = \{X_i\}_{i \in \mathcal{V}}$. For the design of distributed algorithm, we utilize new variables $\bar{Q}_{lq}^{ij}(t)$ with $j \in \mathcal{N}_i^W$, which are the copies of $Q_{lq}^{ij}(t)$. Denote X_c the vector of all variables $\bar{Q}_{lq}^{ij}(t)$, with $i \in \mathcal{V}, j \in \mathcal{N}_i^W$. Rewrite the constraint (2) as

$$Q_{lq}^{ij}(t) + \bar{Q}_{ql}^{ji}(t) = 0, \quad (23)$$

$$Q_{lq}^{ij}(t) = \bar{Q}_{lq}^{ij}(t), \quad \forall (i, j) \in \mathcal{E}^W. \quad (24)$$

Denote the constraint set for X as

$$\Omega_X = \{X | X \text{ satisfies (1), (4), (8) – (10)}\}. \quad (25)$$

For the above set, define the following indicator function as

$$I_X(X) = \begin{cases} 0 & : X \in \Omega_X, \\ +\infty & : \text{otherwise.} \end{cases} \quad (26)$$

Consider the following augmented Lagrangian,

$$\begin{aligned} \mathcal{L}_W(X, X_c, \Lambda_W) &= J_W(X) + I_X(X) \\ &+ \sum_{t \in \mathcal{K}} \sum_{i \in \mathcal{V}} \sum_{l \in \mathcal{J}^i} \sum_{q \in \mathcal{N}_l^{ij}} \left[\lambda_{lq}^{ij}(t) (Q_{lq}^{ij}(t) + \bar{Q}_{ql}^{ji}(t)) \right. \\ &+ \mu_{lq}^{ij} (Q_{lq}^{ij}(t) - \bar{Q}_{lq}^{ij}(t)) + \frac{\rho}{2} (Q_{lq}^{ij}(t) + \bar{Q}_{ql}^{ji}(t))^2 \\ &\left. + \frac{\rho}{2} (Q_{lq}^{ij}(t) - \bar{Q}_{lq}^{ij}(t))^2 \right], \end{aligned} \quad (27)$$

where $\rho > 0$ is a chosen scalar parameter, $\lambda_{lq}^{ij}(t)$ and $\mu_{lq}^{ij}(t)$, $\forall (i, j) \in \mathcal{E}^W, l \in \mathcal{J}^i, q \in \mathcal{N}_l^{ij}$ are called the Lagrange multipliers corresponding to the constraints in (23), (24); and Λ_W is the vector of $\{\lambda_{lq}^{ij}(t)\}$ and $\{\mu_{lq}^{ij}(t)\}$. Next, the variables X, X_c are computed by the following algorithm

$$X^+ \leftarrow \underset{X}{\operatorname{argmin}} \mathcal{L}_W(X, X_c, \Lambda_W), \quad (28a)$$

$$X_c^+ \leftarrow \underset{X_c}{\operatorname{argmin}} \mathcal{L}_W(X^+, X_c, \Lambda_W), \quad (28b)$$

$$\lambda_{lq}^{ij}(t)^+ \leftarrow \lambda_{lq}^{ij}(t) + \rho (Q_{lq}^{ij}(t)^+ + \bar{Q}_{ql}^{ji}(t)^+), \quad (28c)$$

$$\mu_{lq}^{ij}(t)^+ \leftarrow \mu_{lq}^{ij}(t) + \rho (Q_{lq}^{ij}(t)^+ - \bar{Q}_{lq}^{ij}(t)^+), \quad (28d)$$

$$X \leftarrow X^+, X_c \leftarrow X_c^+, \Lambda_W \leftarrow \Lambda_W^+. \quad (28e)$$

Sub-problem (28a) is a quadratic programming and can be solved in finite steps by active set method Cannon et al. (2001). It is worth noting that the sub-problem (28a) can be decomposable to each agent i as

$$X_i^+ = \underset{X_i}{\operatorname{argmin}} \mathcal{L}_{Wi}(X_i, X_c, \Lambda_W), \quad (29a)$$

$$\text{s.t. (1), (3), (5), (10) – (12),} \quad (29b)$$

where

$$\begin{aligned} \mathcal{L}_{Wi}(X_i, X_c, \Lambda_W) &= J_{Wi}(X_i) \\ &+ \sum_{t \in \mathcal{K}} \sum_{l \in \mathcal{J}^i} \sum_{q \in \mathcal{N}_l^{ij}} \left[\lambda_{lq}^{ij}(t) (Q_{lq}^{ij}(t) + \bar{Q}_{ql}^{ji}(t)) \right. \\ &+ \mu_{lq}^{ij} (Q_{lq}^{ij}(t) - \bar{Q}_{lq}^{ij}(t)) + \frac{\rho}{2} (Q_{lq}^{ij}(t) + \bar{Q}_{ql}^{ji}(t))^2 \\ &\left. + \frac{\rho}{2} (Q_{lq}^{ij}(t) - \bar{Q}_{lq}^{ij}(t))^2 \right]. \end{aligned}$$

The problem (29) contains only local variable X_i , then it can be solved locally at each agent i . Note that in the

problem (29), agent i needs to collect $\bar{Q}_{ql}^{ji}(t)$ from agent $j \in \mathcal{N}_i^W$, which is computed from the previous iteration. Thus, the sub-problem (28a) can be solved in a distributed way.

Sub-problem (28b) also can be decomposable to each agent i as follows

$$\bar{Q}_{lq}^{ij}(t)^+ = \underset{\bar{Q}_{lq}^{ij}}{\operatorname{argmin}} \left\{ \begin{aligned} &\mu_{lq}^{ij}(t) (\bar{Q}_{lq}^{ij}(t)^+ - \bar{Q}_{lq}^{ij}) \\ &+ \frac{\rho}{2} (Q_{lq}^{ij}(t)^+ - \bar{Q}_{lq}^{ij})^2 \\ &+ \lambda_{ql}^{ji} (Q_{ql}^{ji}(t)^+ + \bar{Q}_{lq}^{ij}(t)) \\ &+ \frac{\rho}{2} (Q_{ql}^{ji}(t)^+ + \bar{Q}_{lq}^{ij}(t))^2 \end{aligned} \right\}. \quad (30)$$

The above sub-problem has a unique closed-form solution as

$$\bar{Q}_{lq}^{ij}(t)^+ = \frac{1}{2\rho} \left[\mu_{lq}^{ij} - \lambda_{ql}^{ji} + \rho (Q_{lq}^{ij}(t)^+ - Q_{ql}^{ji}(t)^+) \right]. \quad (31)$$

To proceed (31), community i need to collect $Q_{ql}^{ji}(t)^+$ and $\lambda_{ql}^{ji}(t)$ from community $j \in \mathcal{N}_i^W$; these values are computed from the previous steps.

Theorem 1. Denote the set of the optimal solutions of the problem **P1** by S_W^* . Let X^k, X_c^k, Λ_W^k be the update of X, X_c, Λ_W at iteration k in (28) respectively, then we have $\lim_{k \rightarrow \infty} d(X^k, S_W^*) = 0$

Proof: The iterates in (28) are based on ADMM algorithm Boyd et al. (2011), according to part 3.2 of Boyd et al. (2011), $Q_{lq}^{ij}(t) + \bar{Q}_{ql}^{ji}(t) \rightarrow 0$, $Q_{lq}^{ij}(t) - \bar{Q}_{lq}^{ij}(t) \rightarrow 0$ and $\Lambda_W \rightarrow \Lambda_W^*$, where Λ_W^* is a dual optimal point. Thus, from (27), $\mathcal{L}_W(X^k, X_c^k, \Lambda_W^k) \rightarrow J_W(X^k) + I_X(X^k)$ as $k \rightarrow \infty$. Then, from (28a), $X^k \rightarrow \underset{X}{\operatorname{argmin}} \{J_W(X) + I_X(X)\}$, or

$$\lim_{k \rightarrow \infty} d(X^k, S_W^*) = 0.$$

3.2 Solution to **P2**

Let us stack all the local variables of each agent i (except those in X_i) to vector Y_i , and denote $Y = \{Y_i\}_{i \in \mathcal{V}}$. Due to the constraint (15), $P_{ij}(t)$ and $P_{ji}(t)$ need to be computed in the same processor. However, $P_{ij}(t)$ and $P_{ji}(t)$ are held by agents i and j , respectively. To design the distributed solution, at each agent i , we introduce new variables $P_{ij}^c(t)$, which are the copies of $P_{ij}(t)$, $j \in \mathcal{N}_i^E$; and then stack all of them to vector Y_c . The constraint (15) can be rewritten as

$$P_{ij}(t) + P_{ji}^c(t) = 0, \quad (32)$$

$$P_{ij}(t) = P_{ij}^c(t), \quad \forall (i, j) \in \mathcal{E}^E. \quad (33)$$

Denote the constraint set for Y as

$$\Omega_Y = \{Y | Y \text{ satisfies (3) – (7), (11), (12), (13), (14), (16) – (18)}\}. \quad (34)$$

Define the indicator function for the set Ω_Y as

$$I_Y(Y) = \begin{cases} 0 & : Y \in \Omega_Y, \\ +\infty & : \text{otherwise.} \end{cases} \quad (35)$$

We consider the following augmented Lagrangian function

$$\begin{aligned} \mathcal{L}_E(Y, Y_c, \Lambda_E) &= J_E(Y) + I_Y(Y) \\ &+ \sum_{i \in \mathcal{V}} \sum_{j \in \mathcal{N}_i^E} \left[\lambda_{E}^{ij} (P_{ij}(t) + P_{ji}^c(t)) + \frac{\rho}{2} (P_{ij}(t) + P_{ji}^c(t))^2 \right. \\ &\left. + \mu_{E}^{ij} (P_{ij}(t) - P_{ij}^c(t)) + \frac{\rho}{2} (P_{ij}(t) - P_{ij}^c(t))^2 \right], \end{aligned} \quad (36)$$

where $\lambda_E^{ij}(t)$ and $\mu_E^{ij}(t)$, $\forall(i, j) \in \mathcal{E}^E$ are called the Lagrange multipliers corresponding to the constraints in (32),(33); and Λ_E is the vector of $\{\lambda_E^{ij}(t)\}$ and $\mu_E^{ij}(t)$. We propose the following algorithm to solve **P2**

$$Y^+ = \underset{Y}{\operatorname{argmin}} \mathcal{L}_E(Y, Y_c, \Lambda_E), \quad (37a)$$

$$Y_c^+ = \underset{Y_c}{\operatorname{argmin}} \mathcal{L}_E(Y^+, Y_c, \Lambda_E), \quad (37b)$$

$$\lambda_E^{ij}(t)^+ = \lambda_E^{ij}(t) + \rho(P_{ij}(t)^+ + P_{ji}^c(t)^+), \quad (37c)$$

$$\mu_E^{ij}(t)^+ = \mu_E^{ij}(t) + \rho(P_{ij}(t)^+ - P_{ij}^c(t)^+), \quad (37d)$$

$$Y \leftarrow Y^+, Y_c \leftarrow Y_c^+, \Lambda_E \leftarrow \Lambda_E^+. \quad (37e)$$

Similar to the algorithm (28), the updates in (37) also can be decomposable, then we omit the details here for brevity. To proceed (37), each agent i needs to gather $P_{ij}^c(t)$ from agent $j \in \mathcal{N}_i^E$; and each agent i gathers $P_{ji}(t)^+$ and $\lambda_E^{ji}(t)$ from agent $j \in \mathcal{N}_i^E$. Note that these exchanging values are known and available at agent $j \in \mathcal{N}_i^E$ at the current step, then there is no need for global information, and the algorithm (37) is fully distributed.

Theorem 2. Let S_E^* be the set of the optimal solutions to the problem **P2**. Denote the updates of Y, Y_c, Λ_E in (37) by Y^k, Y_c^k, Λ_E^k , respectively. The iterates satisfy $\lim_{k \rightarrow \infty} d(Y^k, S_E^*) = 0$.

Proof: The proof for Theorem 2 is similar to the one for Theorem 1, then it is omitted here for brevity.

4. CASE STUDIES

4.1 Simulation setup

We consider the modified Pescara Network Saldarriaga et al. (2020) as shown in Fig. 3. The Pescara benchmark was first introduced by Bragalli et al. Bragalli et al. (2012) and has been widely used as an example of the water distribution system. The network consists of three water supply plants, 68 pipes, and 64 nodes. To illustrate the effectiveness of the optimization scheme, we divide the network into three communities; each of them is a sub-network with one water supply plant. Furthermore, each community is equipped with three pumps; communities 1 and 2 contain one tank, while community 3 contains two tanks (Fig. 3).

Water and energy can be exchanged between two neighboring communities through the water network and the energy network, respectively. In this part, we assume that the water and energy networks are the same, which implies that any pair of communities that can exchange energy can exchange water as well. We focus on a period of length 24 hours, and the time slot size is 1 hour, then the number of time slots in each period is $K = 24$. The water generation costs vary over the time period, and we assume that the daytime includes $\{k | 7 \leq k \leq 18\}$, and the costs for water generation in the daytime at three water supply plants are chosen as: $C_1(Q_1^G) = 0.001(Q_1^G)^2 + 0.05Q_1^G$, $C_2(Q_2^G) = 0.001(Q_2^G)^2 + 0.05Q_2^G$, $C_3(Q_3^G) = 0.001(Q_3^G)^2 + 0.055Q_3^G$. In the nighttime, we assume that the cost for water generation increases as $C_1(Q_1^G) = 0.0011(Q_1^G)^2 +$

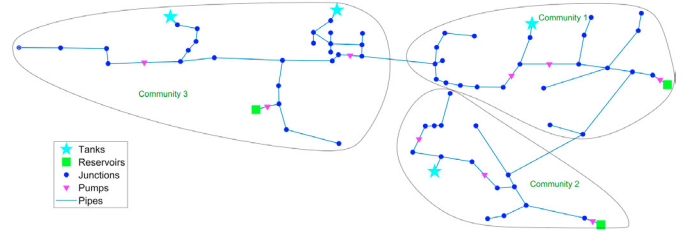


Fig. 3. The modified Pescara water distribution system

$$0.1Q_1^G, C_2(Q_2^G) = 0.0011(Q_2^G)^2 + 0.1Q_2^G, C_3(Q_3^G) = 0.0011(Q_3^G)^2 + 0.22Q_3^G.$$

The water utility at each junction also varies. At the on-peak hours $k \in \{11, 12, 13\}$ (lunchtime) and $k \in \{17, 18, 19\}$ (evening time), the utility functions at all junctions are chosen equally as $U_l^i(D_l^i) = 45D_l^i - (D_l^i)^2$. Other time slots are considered as the off-peak hours, and the utilities are therefore at lower levels as $U_l^i(D_l^i) = 15D_l^i - 0.5(D_l^i)^2$. According to the utility functions, all users are treated equally, but the utility function could be modified to give priority to a certain region. For energy generation, we assume that the cost functions for the three communities are chosen as $E_1(P_1) = 0.005P_1^2 + 0.1P_1$, $E_2(P_2) = 0.005P_2^2 + 0.12P_2$, $E_3(P_3) = 0.005P_3^2 + 0.13P_3$.

4.2 Simulation results

To verify the effectiveness of the proposed optimization algorithms, we consider four cases. In the first case, both water and energy can be exchanged among communities. In Case 2, we only allow water exchange among communities; there is no energy exchange among communities. In Case 3, there is no water and energy exchange among communities. In the last case, similar to Case 1, there are both water and energy exchanges among three communities, but we disable all the tanks of the water network to verify the advantages of the tanks. As shown in TABLE I, exchanging either water or energy among communities can reduce the total cost for society. In Case 4, when we remove all tanks from the water network, the total cost for water and energy generation is higher than in Case 1. This can be explained by the fact that the tanks can reduce the amount of water generation when the cost is high by supplementing water from tanks; then, the total cost is reduced.

Table 1: Water and energy costs

Generation cost (\$)	Water	Energy	Total
Case 1	11042	2845	13887
Case 2	11042	3223	14265
Case 3	11481	3093	14574
Case 4	11406	2653	14059

Next, we present some results in Case 1, in which there are both water and energy exchanges. Figure 4a shows the amount of water generation in the three communities. We can see that during the daytime, the water generation rate in the three communities is higher than during the nighttime. Figure 4b shows the water consumption in three communities. The amount of water consumption is based

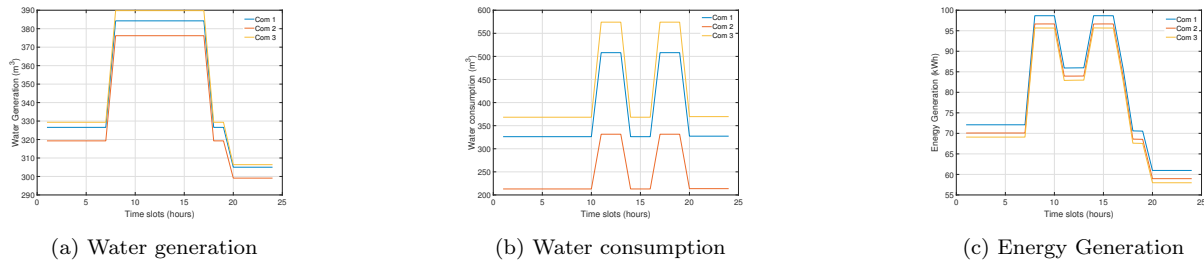


Fig. 4. The results for water generation, consumption, and energy generation in case there are both water and energy exchange

on the utilities at junctions. At peak hours (lunchtime and evening time), junctions require more water; consequently, there is a higher water consumption rate at peak hours (Fig. 4b). The energy generation at three communities is shown in Fig. 4c. During the daytime, there is a larger amount of water generation; thereby, there requires more energy for water distribution as shown in Figure 4c.

5. CONCLUSIONS

This paper presented a distributed optimization method for water coordination with optimal power operation. We first introduced a new model that describes the water and energy network as a multi-agent structure. The water network was divided into several sub-networks, which are called communities of the society. Unlike existing works, we formulated a convex optimization problem, then the convergence of the algorithm could be guaranteed, and we theoretically proved that the updates of the algorithm approach a global solution. The proposed algorithm based on ADMM is fully distributed; eliminating the need for a central coordinator and thereby enhancing the privacy of the system.

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